

Meeting Minutes 10

Project Name: Financial and Operational Analysis of U.S. Hospitals Using Hospital Provider Cost Report Data (2011–2022)

*Meeting Minutes 10A (In-Class – Advanced Modelling & Hyperparameter Tuning Review)
(Multi-Model Comparison & Cross-Validation Strategy Validation)*

Date and Time

24 Mar 2026, 8:00 AM – 9:00 AM

Location

St. Clair College, One Riverside Drive, Room 1004

Attendees

- Professor Manjari Maheshwari
- Manjunath Muthineni (0878795)
- Rajesh Thota (0873782)
- Neha (0873038)
- Abhishekh Choudhary (0873460)
- Krishna Chaitanya Venuturumilli (0874277)

Absent

None

Agenda

- Present advanced ensemble and boosting models
- Review GridSearchCV hyperparameter tuning approach
- Validate cross-validation strategy
- Identify and eliminate data leakage

Discussion Points

- Manjunath presented the implementation of multiple advanced machine learning models including Random Forest, Gradient Boosting, HistGradientBoosting, ExtraTrees, and a Stacking ensemble. All models were developed using sklearn pipelines to ensure a consistent and reproducible preprocessing workflow across experiments.
- Rajesh provided a detailed explanation of the GridSearchCV hyperparameter tuning process. He described how parameter grids were carefully designed to balance computational efficiency and model

performance. Key parameters tuned included `n_estimators`, `max_depth`, `learning_rate`, and `min_samples_split`.

- Neha elaborated on the evaluation framework, emphasizing the use of 3-fold cross-validation during hyperparameter tuning and 5-fold cross-validation for final model evaluation. This approach ensured that performance metrics were stable and not dependent on a single data split.
- Abhishekh highlighted the importance of using **cross-validated R^2 (CV R^2)** as the primary performance metric. He explained that CV R^2 provides a more reliable estimate of model generalization compared to a single train-test split, which can lead to inflated or misleading results.
- Krishna conducted a comprehensive data leakage audit and identified variables that were directly derived from target metrics, such as financial ratios and aggregated cost components. These variables were removed from the feature set to prevent the model from learning trivial relationships. Additionally, high-cardinality identifiers such as `fips` and `medicare_cbsa_number` were excluded to avoid overfitting through geographic memorization.
- Professor Maheshwari reviewed the entire modeling pipeline and strongly endorsed the team's methodology. She emphasized that eliminating leakage and using cross-validation are critical for ensuring that the model performs reliably in real-world scenarios. She also cautioned against over-reliance on single-split evaluation results.

Decisions Made

- Cross-validated R^2 (CV R^2) formally adopted as the **primary model evaluation metric**.
- Data leakage handling strategy approved and finalized.
- Ensemble models confirmed as the most suitable approach for the dataset.
- Random Forest and ExtraTrees selected as the leading candidate models for final deployment.

Action Items

- Finalize model comparison table using cross-validated results.
- Develop a clean and reproducible final training notebook.
- Ensure all preprocessing steps are standardized across pipelines.

Next Meeting Date and Time

26 Mar 2026 (Group meeting – ANN evaluation & model finalization)

Minutes Prepared By

Manjunath Muthineni

Meeting Minutes 10B (Group Meeting – Neural Network Evaluation & Model Finalization)
(ANN Performance Assessment & Final Model Selection)

Date and Time

26 Mar 2026, 7:00 PM – 9:00 PM

Location

Online (Microsoft Teams)

Attendees

- Manjunath Muthineni (0878795)
- Rajesh Thota (0873782)
- Neha (0873038)
- Abhishekh Choudhary (0873460)
- Krishna Chaitanya Venuturumilli (0874277)

Absent

None

Agenda

- Evaluate Artificial Neural Network (ANN) performance
- Compare ANN with ensemble models
- Finalize model selection

Discussion Points

- Manjunath developed an Artificial Neural Network (ANN) using TensorFlow/Keras, consisting of two hidden layers with ReLU activation and dropout regularization. The model was integrated into the sklearn ecosystem using scikeras, enabling compatibility with GridSearchCV for systematic tuning.
- Rajesh conducted hyperparameter tuning across multiple configurations, including variations in neuron counts, dropout rates, optimizers (adam, rmsprop), batch sizes, and training epochs. This ensured that the ANN was evaluated under optimized conditions rather than default settings.
- Neha presented the evaluation results, showing that the ANN achieved lower cross-validated R^2 scores compared to ensemble models. While the ANN captured nonlinear relationships, its performance did not exceed that of Random Forest and ExtraTrees.

- Abhishekh discussed the practical implications of using ANN models, highlighting increased computational cost, longer training time, and the requirement for additional preprocessing such as feature scaling. He emphasized that these factors introduce complexity without providing a corresponding performance benefit.
- Krishna concluded that tree-based ensemble models are inherently better suited for structured tabular data, as they can capture nonlinear relationships and feature interactions without requiring extensive preprocessing or large datasets.

Decisions Made

- ANN excluded from the final production model set due to lower performance and higher complexity.
- Final models confirmed:
 - Random Forest (Profit Margin prediction)
 - ExtraTrees (Charity Care Ratio prediction)

Action Items

- Build the final model training notebook with only the selected models.
- Begin development of the application interface for model deployment.
- Document ANN methodology and results for inclusion in the project report appendix.

Next Meeting Date and Time

31 Mar 2026 (In-class – final model review & deployment discussion)

Minutes Prepared By

Krishna Chaitanya Venuturumilli